BITCOIN PRICE FORECASTING

Data Science Course Project — Educational Pilot Study

Created in 2024 · Portfolio Edition

Victor Buret

AI & Automation Builder · Python · SQL · n8n

A transparent learning project about historical time-series modelling. It is not a trading system and does not provide financial advice.

Project at a glance

01	**Purpose:** explore a historical five-day-ahead Bitcoin price forecasting task.
02	**Type:** final Data Science course project and educational pilot study.
03	**Original scope:** historical market data, selected on-chain indicators, and model comparison.
04	**Portfolio boundary:** records the learning work honestly without presenting it as a live financial product.

Research question

01 Can selected historical market and on-chain variables be used to explore a five-day-ahead Bitcoin price forecasting task?

02 The project investigates the modelling question; it does not recommend buying, selling, or holding any asset.

Original data scope

01 Historical Bitcoin-related data and selected on-chain indicators were used through January 2024.

02 Selected market data were requested from 2015 through January 2024.

03 The original JSON inputs and generated finance CSV are not included in this public portfolio edition.

Approaches explored

01	Linear regression and nearest-neighbour regression.
02	Tree-based ensembles and gradient-boosting approaches.
03	Neural-network approaches, including MLP and Keras Sequential models.
04	SARIMAX time-series modelling.

What the course project demonstrates

01	Data acquisition, cleaning, merging, and exploratory analysis.
02	Feature engineering for a time-dependent target.
03	Comparison of several modelling families.
04	A practical understanding of why evaluation design matters in time-series work.

Important limitations

01 Some transformations were created before the temporal split, which can introduce information leakage.

02 Generic cross-validation is not sufficient for a time-series forecasting claim.

03 Historical metrics in the source notebook include transformed values and are not presented here as current real-price accuracy.

04 The original study ends in January 2024; it does not establish present-day performance.

A responsible methodological refresh

01	Keep data preparation inside each training window.
02	Use walk-forward validation or TimeSeriesSplit.
03	Retain a final untouched holdout period.
04	Report errors in original price units as well as transformed scale where applicable.
05	Record retrieval dates, source boundaries, and exact package versions.

Portfolio interpretation

01 This project is evidence of coursework in data preparation, exploratory analysis, model comparison, and critical evaluation.

02 It is not evidence of profitable trading performance, investment suitability, or a production-ready forecasting service.

Reproducibility status

01	The public notebook is provided as an English, output-free source record.
02	A reproducible rerun requires documented data retrieval, a controlled environment, and time-aware evaluation.
03	Downloaded market data should remain outside version control unless its license expressly permits publication.

Financial disclaimer

01	Cryptocurrency markets are volatile.
02	Nothing in this project is financial advice, investment advice, or a recommendation to buy, sell, or hold an asset.
03	Any future work should report limitations and uncertainty transparently.